

1.2.47

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Problem

Evaluate:

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin^2(2x)}{\left(x - \frac{\pi}{2}\right)^2} \quad (1)$$

Solution

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin^2(2x)}{(x - \frac{\pi}{2})^2} \quad (2)$$

$$\frac{0}{0} \text{ form} \quad (3)$$

So, we use L'Hopital's Rule :

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad (4)$$

Differentiate:

$$\frac{d}{dx} [\sin^2(2x)] = 4 \sin(2x) \cos(2x) \quad (5)$$

$$\frac{d}{dx} [(x - \frac{\pi}{2})^2] = 2(x - \frac{\pi}{2}) \quad (6)$$

Solution

$$\frac{2 \sin(2x) \cos(2x)}{x - \frac{\pi}{2}} \quad (7)$$

We know that:

$$2 \sin(2x) \cos(2x) = \sin(4x) \quad (8)$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(4x)}{x - \frac{\pi}{2}} \quad (9)$$

Again $\frac{0}{0}$ form

So, differentiate again

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} 4 \cos(4x) \quad (10)$$

$$\Rightarrow 4 \quad (11)$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin^2(2x)}{\left(x - \frac{\pi}{2}\right)^2} = \boxed{4} \quad (12)$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Define the function
def f(x):
    return (np.sin(2*x)**2) / ((x - np.pi/2)**2)

# Create x values avoiding division by zero
x = np.linspace(-2*np.pi, 2*np.pi, 1000)
x = x[x != np.pi/2]

# Compute y values
y = f(x)

# Plot the function
plt.figure(figsize=(10, 5))
plt.plot(x, y, label=r $f(x) = \frac{\sin^2(2x)}{(x - \frac{\pi}{2})^2}$ )
```

Python Code

```
plt.scatter([np.pi/2], [4], color='red', zorder=5, label=Point (
    /2, 4))

# Horizontal dashed line for limit
plt.axhline(4, linestyle='--', color='red', label=limit = 4)

# Labels and title
plt.title(Broad View of f(x) with Oscillations)
plt.xlabel(x)
plt.ylabel(f(x))

# Grid and legend
plt.grid(True)
plt.legend()

# Show plot
plt.show()
```

Plot

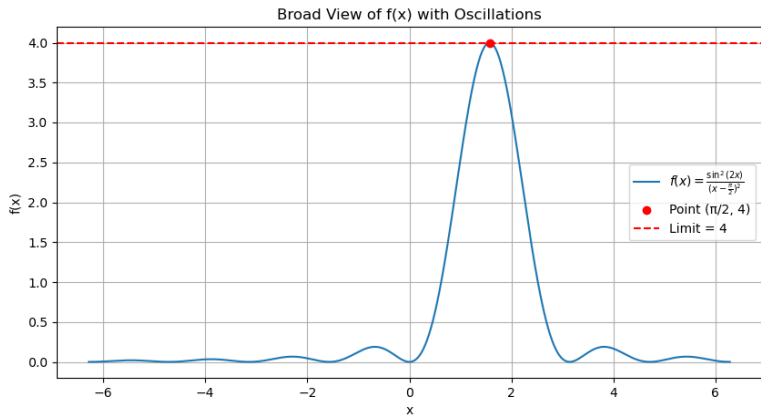


Figure: Plot